

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Formative Assessment 1

Formative Assessment (FA-1)

Class: Class 8

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Functions

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. What is the output of the following Python code?

```
def greet(name):  
    return "Hello, " + name  
  
print(greet("Alice"))
```

- "Hello, Alice"
 - "Hello, "
 - "Alice"
 - Error
- b. Which keyword is used to define a function in Python?
- func
 - define
 - def
 - function
- c. A function that returns a value is called a:
- Void function
 - Pure function
 - Generator function
 - Lambda function
- d. What is the default return value of a Python function that does not have a return statement?
- 0
 - False
 - None
 - Empty string
- e. In Python, which of the following is used to call a function?
- ()

g. []

h. {}

Section II: Short Answer (5 x 2 = 10 marks)

- Define a function. Give its syntax in Python.
- Differentiate between **arguments** and **parameters** with one example each.
- Write the output of:

```
def factorial(n):  
    if n == 0:  
        return 1  
    return n * factorial(n - 1)  
  
print(factorial(5))
```

- Explain the difference between a **local variable** and a **global variable** inside a function.
- What is the purpose of the return statement in a function? Give an example.

Section III: Long Answer (5 x 1 = 5 marks)

- Write a Python function `is_prime(n)` that checks whether a given number `n` is a prime number or not. Include docstring, proper indentation, and handle edge cases. Call the function for 5 test values and show the output.

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Defines a function `calculate_bmi(weight_kg, height_m)` that computes Body Mass Index
- Includes input validation (positive numbers only)
- Categorises the BMI: Underweight (< 18.5), Normal (18.5-24.9), Overweight (25-29.9), Obese (>= 30)
- Calls the function for at least 4 different persons and displays a formatted table of results

Part C: Notebook Maintenance (5 marks)

- Neatness and organisation of notebook (2 marks)
- Completion of all assigned classwork and homework (2 marks)
- Proper headings, dates, and indexing (1 mark)